User Guide

Delphi App Translation Studio

Version 1.0

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Windows • Delphi VCL and FireMonkey • Win32 and Win64

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1. Welcome

Delphi App Translation Studio is an offline-first Windows developer tool for adding localization to existing Delphi VCL and FireMonkey applications. It scans designer-authored forms and Delphi resourcestring declarations, stores the results in an editable development catalog, automatically translates unresolved text through Google Cloud Translation or DeepL, validates translation safety, and exports a compact JSON pack for the target application.

The Studio itself is built with FireMonkey, but it works with both VCL and FMX target projects. The supported build targets are Windows Win32 and Win64. macOS, iOS, Android, Linux, C++Builder, and runtime cloud translation are outside the product scope.

> **Offline by design.** Only the developer's Studio computer needs Internet access while creating translations. The finished target application reads local JSON language packs and never needs Internet access, a provider account, or an API key.

1.1 What the Studio changes

- Scanning is read-only and does not alter the selected project.
- Saving creates project-local Localization\Development catalog files.
- Export creates project-local Localization\Languages runtime packs.
- Component Integration generates a setup kit under the Studio export folder and does not write to the selected target project.
- The developer places one manager on the primary form in Delphi; those normal IDE-authored form and uses-clause changes remain visible in source control.
- Automatic Source Integration remains an explicit advanced fallback with preview, verified backup, apply, and restore.

2. Installation and Startup

The open-source project can be opened and compiled by anyone with Delphi 13 / RAD Studio 13 Florence. No installer is required for development builds. Open DelphiAppTranslationStudio.dproj, choose Win32 or Win64, and build.

Build products are placed under bin\<Platform>\<Configuration>. Compiler units are placed under dcu. The project deliberately does not target mobile or macOS platforms.

2.1 First launch

1. Run DelphiAppTranslationStudio.exe.

2. Confirm that the title reads Delphi App Translation Studio.
3. The Studio opens maximized. Use the left workflow panel; every page expands into the available workspace and the blue selection bar follows Project, Scan, Translate, Validation, Export, Integration, and Provider Settings.

2.2 Install the localization components

1. Close RAD Studio. In the Translation Studio, open Integration, leave Component Integration selected, and choose Install / Repair Components. The same operation is available by double-clicking tools\Install-DATLanguageManagerComponents.cmd, which uses ExecutionPolicy Bypass for the signed-off project script.
2. Approve the Windows elevation prompt. The installer builds and tests the packages, copies the matching Win32 BPLs to the public RAD Studio Bpl folder, preserves Delphi's complete installed Embarcadero design-package registry baseline, excludes stale optional registrations whose BPL files are absent, and registers the self-contained DAT design BPL for the current user.
3. Restart RAD Studio. Open Component > Install Packages and confirm the normal Embarcadero packages are still listed, including Embarcadero FMX Components for FMX work, and that the DAT package is checked. Then open a form and confirm DAT Localization appears in the Tool Palette with the language manager and language combo box.
4. If automation cannot be used, build the matching Win32 design package and use Component > Install Packages > Add to select the compiled .bpl. Never choose Component > Install Component and never select a .dpk in that wizard; a .dpk is package source, not the installable artifact.
5. Runtime BPLs are also copied for projects that use runtime packages. A normally linked application compiles the generated kit's ComponentSource units into the executable.

> **Explicit, automated installation.** The installer is developer-invoked, requires RAD Studio to be closed, and reports the destination and result. It never alters a target application.

3. The End-to-End Workflow

Step	Purpose	Primary output
1 Project	Open and identify a Delphi project.	Project profile
2 Scan	Extract designated designer text and resourcestrings.	Scan result

Step	Purpose	Primary output
3 Translate	Select a language and translate unresolved text automatically.	Saved machine-translation catalog
4 Validation	Check completeness and structural safety.	Issue list
5 Export	Create the compact offline pack.	Runtime JSON
6 Integration	Generate a non-mutating component setup kit; optionally use advanced automatic integration.	Component kit or advanced preview
7 Provider Settings	Configure and test Google Cloud Translation or DeepL.	Secure provider configuration

4. Open and Scan a Project

1. Choose Project and select Open Delphi Project.
2. Open a .dproj when available; a .dpr is also accepted.
3. Review the detected framework, platforms, form count, and source count.
4. Choose Scan. The Studio reads text DFM/FMX resources and resourcestring declarations.
5. Review the entry count, elapsed milliseconds, breakdown, and result list.

> **Binary forms.** The scanner expects text DFM/FMX resources. If a legacy DFM is binary, use Delphi's normal form conversion facilities before scanning and keep a source-control copy.

4.1 Text that is collected

- Captions, Text values, prompts, hints, headings, and designated list content.
- Memo and string-list content when it represents user-visible text.
- Delphi resourcestring symbols with stable unit-and-symbol keys.
- Locale metadata such as date, time, number, and currency formatting is entered on the Languages page rather than guessed from UI text.

5. Create and Edit a Language Catalog

1. Choose Translate after scanning.

2. Select the source language, normally English (United States) [en-US].
3. Select the target language from the built-in language list. The Studio records its locale code and native name.
4. Review the automatically selected left-to-right or right-to-left direction.
5. Review or edit the date, time, decimal, thousands, and currency fields. Blank fields are initially populated from Delphi TFormatSettings for the target locale.
6. Choose Save.
7. Click the blue catalog path to select the saved JSON file in File Explorer.

The catalog is saved under Localization\Development using the project name and target locale. Each target language has its own development catalog. Existing translations are preserved on later scans when the stable key and source text remain unchanged. Changed source text is flagged for review, and removed entries become obsolete rather than disappearing silently.

5.1 Translation statuses

Status	Meaning
Needs translation	No usable target text is present.
AI draft	Legacy provenance retained when opening a catalog created by an earlier Studio build.
Machine translated	DeepL or Google produced a draft that requires review.
Imported	CSV supplied target text that requires review.
Edited	A person changed or entered the target text.
Reviewed	A person reviewed the target text.
Approved	The entry is release-approved.
Source changed	The source changed after a translation existed.
Excluded	The entry is intentionally omitted.
Obsolete	The source entry no longer exists in the latest scan.
Error	The entry needs correction after a failed operation or check.

5.2 Automatic Google or DeepL workflow

1. Open Provider Settings and select Google Cloud Translation or DeepL.
2. Paste the provider API key into the masked field, choose secure Windows Credential Manager storage or session-only use, and select Replace / Save Key.
3. Choose Test Connection. Resolve any authentication, billing, restriction, firewall, or quota error before translating a project.
4. Return to Translate, select the target language, and choose Translate Automatically.
5. Confirm the displayed provider and unresolved-entry count. The Studio sends only eligible unresolved source strings; Reviewed, Approved, Excluded, and Obsolete entries are not replaced.
6. The Studio processes provider requests in bounded batches, records each result as Machine translated with Google or DeepL provenance, saves the development catalog automatically, and leaves every result ready for human review.

No extra software installation. Automatic translation is built into the Studio. It requires only a supported provider account and API key; no command-line AI agent, Python package, browser extension, or separate translation utility is required.

5.3 Safety, context, and focused review

- Designer-property entries are applied automatically by the VCL or FMX runtime adapter.
- Pascal resourcestring entries require an explicit TranslateText(Key, Fallback) call at the intended code location. Mark Manual wiring confirmed only after adding and reviewing that call.
- Translation origin is independent of Draft, Reviewed, and Approved status. A provider result is never silently approved.
- Only Needs Translation, Source Changed, and Error entries are eligible. Existing Machine-translated, Imported, Edited, Reviewed, Approved, Excluded, and Obsolete work is protected unless the developer changes its status deliberately.
- The terminology profile records application context, audience, tone, formality, protected names, and preferred terminology.
- Validation focuses on placeholders, accelerators, inconsistent repeated terms, source changes, and runtime wiring.
- Exact-source translations from other stable keys are suggestions only. The developer must explicitly accept one, and the accepted target remains Edited rather than inheriting approval.

6. Provider Settings and Secure Key Storage

Provider Settings supports DeepL API Free, DeepL API Pro, and Google Cloud Translation Basic v2. Provider accounts, billing, quotas, prices, and supported languages are controlled by the provider and may change.

A remembered key is stored as a Windows Generic Credential for the current Windows user. The key is not written to the Delphi project, JSON catalogs, exported packs, source distribution, or Git repository. Session-only keys are cleared when the Studio closes.

Control	Behavior
Provider	Select DeepL or Google Cloud Translation.
DeepL API plan	Select API Free or API Pro so the Studio uses the matching endpoint.
API key	Masked field for a new or replacement key.
Remember securely	Stores the key as a Windows Generic Credential.
Session only	Clear Remember before saving; the key is held only until shutdown.
Replace / Save Key	Records the entered key using the selected storage choice.
Test Connection	Sends a small English-to-Italian request.
Remove Key	Deletes stored and session copies for the selected provider.
Timeout / batch	Controls 5-300 second requests and 1-50 strings per request.

7. Obtain and Enter a DeepL API Key

> **Use a DeepL API account.** A normal DeepL Translator subscription is not automatically a DeepL API plan. Confirm that the account includes API Free or API Pro.

1. Open the official DeepL developer site at https://developers.deepl.com/ and review current API account requirements.
2. Create or sign in to a DeepL account and subscribe to DeepL API Free or DeepL API Pro as appropriate.

3. Open the account's API Keys tab. Create a separate key for this Studio when the account interface permits multiple keys.
4. Copy the key once and keep it out of source code, JSON catalogs, issue trackers, email, and screenshots.
5. In the Studio, choose Provider Settings and select DeepL.
6. Select API Free or API Pro to match the account. Free uses api-free.deepl.com; Pro uses api.deepl.com.
7. Paste the key into the masked field.
8. Leave Remember securely checked for Windows Credential Manager, or clear it for Use for This Session Only.
9. Choose Replace / Save Key, then Test Connection.
10. If the test fails, verify the plan selection, key status, account quota/billing, firewall, and target-language availability.

DeepL key and authentication reference: https://developers.deepl.com/docs/getting-started/auth

DeepL quickstart: https://developers.deepl.com/docs/getting-started/quickstart

DeepL multiple-key guidance: https://developers.deepl.com/docs/multiple-api-keys

8. Obtain and Enter a Google API Key

> **Google Basic v2 only.** The Studio uses Cloud Translation Basic v2 because it supports API-key authentication. Cloud Translation Advanced v3 uses different authentication and is not implemented.

1. Open Google Cloud Console at https://console.cloud.google.com/ and sign in with the account that will own billing and credentials.
2. Use the project selector to create a dedicated project, such as Delphi App Translation Studio, or select an existing project whose billing and key lifecycle you control.
3. Open Billing for that project and link an active billing account. Cloud Translation setup requires billing to be enabled even when usage might fall within a current no-charge allowance.
4. Open APIs & Services, then Library. Search for Cloud Translation API, open it, and choose Enable. Confirm that the selected project is still the intended project.
5. Open APIs & Services, then Credentials. Choose Create credentials, then API key. Create a standard API key; do not bind it to a service account for this Studio.
6. Name the key clearly, for example Delphi App Translation Studio - <computer or developer>. Google requires at least one API restriction when a key is created in the console.
7. Under API restrictions, choose Restrict key and select only Cloud Translation API (service translate.googleapis.com), then save. Do not leave the key usable with every Google API.
8. Application restrictions are recommended by Google, but the available types are websites, server IP addresses, Android, and iOS. A native Windows desktop application has no matching signed-

application restriction. Use an IP-address restriction only when the developer computer exits through a stable public IP that you control; do not select website, Android, or iOS restrictions for the Studio.

9. Copy the displayed key string immediately and keep it out of source code, JSON, screenshots, email, issue trackers, and chat. The key ID or display name is not the usable key string.
10. Review Cloud Translation pricing, quotas, and billing budget alerts before a large project. API keys associate requests with the project for quota and billing.
11. In the Studio, choose Provider Settings and select Google Cloud Translation. The DeepL plan list becomes disabled because it does not apply.
12. Paste the key into the masked field. Leave Remember securely checked to store it as a Windows Generic Credential, or clear the check box for session-only use.
13. Choose Replace / Save Key, then Test Connection. A successful test translates one small English phrase to Italian.
14. If the test returns HTTP 403, verify the selected project, billing, API enablement, and API/application restrictions. HTTP 429 normally indicates a quota or rate limit. After changing a Google restriction, allow a few minutes for propagation and test again.

Google Cloud Translation setup: https://docs.cloud.google.com/translate/docs/setup

Google API-key creation and management: https://docs.cloud.google.com/docs/authentication/api-keys

Google API-key restrictions: https://docs.cloud.google.com/api-keys/docs/add-restrictions-api-keys

Cloud Translation authentication: https://docs.cloud.google.com/translate/docs/authentication

9. Direct Provider Translation

1. Open or create the target-language catalog.
2. Confirm the source and target language codes.
3. Configure and test a provider under Provider Settings.
4. Choose Translate Automatically on the Translate page.
5. Read the confirmation message showing how many unresolved strings will be sent to the provider. Cross-key suggestions are never accepted automatically.
6. Choose Yes to begin. Existing complete reviewed or approved work is preserved.
7. Wait for all batches. Transient HTTP 429 and provider server failures receive bounded retries.
8. Review every machine-translated entry for meaning, placeholders, accelerators, product terminology, tone, and available control space.
9. The Studio saves the catalog automatically. Run Validation after reviewing the results.
10. For focused work, Review and Approve the selected entry. For a large catalog, Review All and Approve All provide separate, confirmed catalog-wide decisions and save once after each operation.

Cost control. The confirmation count is not a price quote. Provider billing can depend on characters and account terms. Check the provider dashboard and quotas before a large request.

10. Validate and Export

Structural validation checks catalog metadata, required translations, duplicate keys, source changes, Delphi indexed/sequential placeholders, accelerator keys, and excluded/obsolete conditions. Errors block export. Manual resourcestring wiring is reported as a runtime-readiness warning. Linguistic Reviewed and Approved states are reported separately and are not inferred from structural validation.

1. Choose Validation and Run Validation.
2. Read the summary: errors block export, warnings request review, and information messages require no action. Double-click an entry-specific issue to open that catalog entry on Translate.
3. Repeat until no errors remain.
4. Choose Export and Export Runtime Pack.
5. Record the output path under Localization\Languages.

11. Integrate a Target Application

Component Integration is the recommended path. It generates validated JSON packs, an English source pack, the applicable component/runtime source, a machine-readable manifest, deployment script, and exact setup instructions. The Studio writes these files only under export\component-integration and never opens the selected target project for writing.

The target application remains completely offline. Provider code and API keys are never added to it. One manager on the primary form supervises the application; ordinary forms do not need individual components.

1. Select Integration and leave Integration method set to Component Integration (Recommended).
2. Choose Build Integration Plan. Confirm the detected framework and translated-pack count.
3. Choose Generate Component Kit. Select README.txt and the generated manifest/files in the read-only inspection panes.
4. If DAT Localization is not already in the Tool Palette, choose Install / Repair Components, close RAD Studio if prompted, complete the installer, and restart RAD Studio.
5. Open the target application's primary form in the Delphi Form Designer and place one TDATVCLLanguageManager or TDATFMXLanguageManager from DAT Localization.
6. Set ApplicationId exactly to the detected Delphi project name. Leave LanguagesFolder as Localization\Languages and set SourceLanguage to the catalog source locale, normally en-US.

7. Optionally place TDATVCLLanguageComboBox or TDATFMXLanguageComboBox and set its LanguageManager property to the manager. Size and align it in the designer like every other visual control.
8. Add the kit's ComponentSource folder to the target Search Path, or reference the installed component source location. Delphi will persist the manager field and component unit through normal designer operations.
9. Copy the kit's Localization folder beside every built Win32/Win64 executable, or run Deploy-LanguagePacks.ps1 with that executable directory.
10. Build and run the target. The saved language is applied during startup. Choosing another locale from the optional selector immediately retranslates open forms and saves the preference.

11.1 Component properties and behavior

Property or control	Purpose
ApplicationId	Must match runtime-pack applicationId; normally the Delphi project name.
LanguagesFolder	Pack folder relative to the executable unless an absolute path is supplied.
SourceLanguage	Source locale used when no translated pack is selected.
AutoLoadPreferred	Loads the saved per-user locale during manager initialization.
AutoTranslateOwner / AutoTranslateNewForms	Applies the active pack to the primary and later forms.
ReapplyOpenForms	Immediately retranslates open forms after a language change.
PreserveControlState	Protects writable user data, focus, selections, and list selection while text changes.
FormIdentityMappings	Optional FormClass=ScannerRoot mappings for renamed or inherited forms.
Language combo box	Populates from validated packs and calls the linked manager without taking over OnChange.

11.2 Runtime files and preferences

- Deploy JSON packs beside the target executable under Localization\Languages.
- Deploy-LanguagePacks.ps1 and manual copying support custom executable output layouts.
- The selected language is stored in %LOCALAPPDATA%\<ApplicationId>\language.ini.
- The executable folder can therefore remain read-only, as it normally is under Program Files.
- FMX uses additive before-show/release messages. VCL uses additive application idle/modal notifications and direct owner application.
- Selecting or reselecting a language applies the selected pack to every open form immediately. English is a real generated pack, so returning from another language restores all scanned source strings without restarting.

11.3 Automatic Source Integration (Advanced)

Use the advanced mode only when the component path is unsuitable and after committing or otherwise protecting the target project. It retains the previous generated-unit, menu-resource, preview, verified backup, atomic apply, build/deploy, restore, and Complete Reset workflow.

1. Change Integration method to Automatic Source Integration (Advanced).
2. Enter the designer language-menu component name and build the plan.
3. Generate the exact preview and inspect any file that needs closer review.
4. Authorize once, optionally select build/deploy, and choose Apply.
5. Use Restore for the session backup or Complete Reset for a recorded pre-integration baseline.

12. Translate the Studio Itself

1. Run the English Studio and select DelphiAppTranslationStudio.dproj.
2. Scan it and create the desired language catalog.
3. Translate, review, validate, and export the pack.
4. Copy the exported pack to the Studio project or deployed executable's Localization\Languages folder.
5. Add the locale to the designer-authored Studio Language menu when a selectable menu item is required.
6. Close the running Studio, rebuild when the menu changed, and start the executable.
7. Select the new language. Open forms change immediately and the preference is retained for the next launch.

The running executable is never overwritten. The JSON pack and preference are separate from the executable, and the rebuild supplies the updated menu on the next run.

13. Troubleshooting

Symptom	Likely action
Project framework unknown	Open the .dproj, confirm VCL/FMX units and project metadata, then rescan.
No scan entries	Confirm text DFM/FMX resources and designated user-visible properties.
HTTP 401/403	Replace the key; verify provider plan, API enablement, restrictions, and billing.
HTTP 429	Quota or rate limit reached; wait, reduce batch size, or review provider quota.
DeepL test fails only on one plan	Select API Free or API Pro to match the account endpoint.
Google test fails	Confirm Basic v2 Cloud Translation API is enabled and key API restriction permits it.
Export blocked	Run Validation and correct every error.
Language selector is empty	Confirm valid nonempty packs, matching applicationId, and the deployed Localization\Languages folder; call RefreshLanguages after late deployment.
Component missing from Tool Palette	Install the matching Win32 design BPL and keep its runtime dependencies available.
Preference cannot be found	Check %LOCALAPPDATA%\<ApplicationId>\language.ini, not the executable folder.
Target remains English	Confirm manager ApplicationId, JSON applicationId/locale, deployment folder, preference, and manager initialization errors.

14. Security and Privacy

- Only confirmed source strings are sent to the selected provider.
- Do not send secrets, personal data, customer data, or regulated information as UI text.
- Remembered keys are Windows Generic Credentials; session keys are not persisted.
- Removing a key affects only the selected provider.
- Provider settings JSON contains no API key.
- Catalogs, runtime packs, deployment packages, logs, backups, and Git should contain no key.
- Rotate a key immediately in the provider console if it may have been exposed, then replace it in the Studio.

15. File and Folder Reference

Location	Contents
Localization\Development	Editable full development catalogs.
Localization\Languages	Compact offline runtime JSON packs.
export	Generated previews and temporary product output.
export\component-integration	Recommended non-mutating component setup kits.
packages\runtime / packages\design	Core/framework runtime packages and VCL/FMX IDE packages.
docs\guides / docs\pdf	Editable guides and companion PDFs.
%LOCALAPPDATA%\DelphiAppTranslationStudio	Studio settings and language preference, but no remembered key.
Windows Credential Manager	Remembered DeepL/Google Generic Credentials.
%LOCALAPPDATA%\<ApplicationId>	Integrated application's selected-language preference.

16. Provider Reference and Change Notice

Provider setup screens and commercial terms can change after this guide is published. Before creating a production key, compare these steps with the official pages listed in Chapters 7 and 8. Prefer a dedicated, restricted key and review the provider dashboard after the first bulk run.

This guide documents the implemented Studio as of August 8, 2026.